

Quantum information processing is the result of using the physical reality that quantum theory tells us about for the purposes of performing tasks that were previously thought impossible or infeasible

- amount of resources used by a computer to solve a problem

• time, space

- depends on the physical architecture of the computer

$$2n^3 + 3 \rightarrow O(n^3)$$

$$4n^2 + n + 7 \rightarrow O(n^2)$$

$$O(f(n)) \rightarrow \text{upper bound}$$

$$O(f(n)) \rightarrow \text{lower bound}$$

$$O(n^k) \text{ polynomial}$$

$$O(n) \text{ linear}$$

$$O(\log n) \text{ logaritmico}$$

$$O(2^n) \text{ exponencial}$$

Church-Turing Thesis

- computing problem can be solved on any computer if and only if it can be solved on a Turing Machine

a function is computable by a TM if and only if it is computable by some realistic computing device

- a probabilistic Turing machine can efficiently simulate any realistic model of computation

It is possible to describe special-purpose (classical) machines for solving certain problems in such a way that a probabilistic TM simulation may require an exponential overhead in time or space

a model of computation which is consistent with the laws of physics

we explicitly account for all the physical resources used

- classical physics is not powerful enough to efficiently simulate quantum physics

Quantum Strong Church-Turing Thesis

- a quantum Turing machine can efficiently simulate any realistic model of computation

Uniform families of reversible circuits (model of computation)

carry bit values to gates

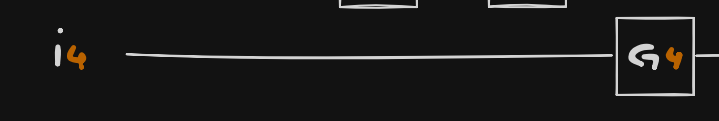
elementary operations

acyclic circuits (linear fashion) the wires never feed back to a prior location

$C_n \rightarrow$ circuit with n wires

$G \rightarrow$ gate

C_4 :



family of circuits $\{C_n \mid n \in \mathbb{Z}^+\}$

uniform if we can easily construct each C_n cannot sneak computational power into the definitions of the circuits themselves

circuits can be generated by a Turing machine in time $O(n^k |C_n|)$

universality A set of gates is universal for classical computation if, for any positive integers n, m and function $f: \{0,1\}^n \rightarrow \{0,1\}^m$, a circuit can be constructed for computing f using only gates from that set

$\{ \text{NAND}, \text{FANOUT} \}$

complexity

number of gates used in the circuit $|C_n|$

depth of the circuit

space

must be fault-tolerant

need a computing model capable of simulating arbitrary 'realistic' physical and quantum devices

quantum version of the strong Church-Turing Thesis

probabilistic Turing machine

quantum computing model

single bit state 0 with probability p_0

state 1 with probability p_1

both deterministic and probabilistic circuits

$$\text{NOT} \begin{pmatrix} p_0 \\ p_1 \end{pmatrix} = \begin{pmatrix} p_1 \\ p_0 \end{pmatrix}$$

$$\text{NOT} \begin{pmatrix} q_0 \\ q_1 \end{pmatrix} = \begin{pmatrix} q_1 \\ q_0 \end{pmatrix}$$

$$\text{NOT} \begin{pmatrix} p_0 \\ p_1 \end{pmatrix} = \begin{pmatrix} q_0 \\ q_1 \end{pmatrix}$$

two bits: first bit state 0 with probability p_0

state 1 with probability p_1

second bit state 0 with probability q_0

state 1 with probability q_1

$$\text{prob}(ij) = p_i q_j$$

$$\begin{pmatrix} p_0 q_0 \\ p_0 q_1 \\ p_1 q_0 \\ p_1 q_1 \end{pmatrix} = \begin{pmatrix} p_0 \\ p_1 \end{pmatrix} \otimes \begin{pmatrix} q_0 \\ q_1 \end{pmatrix}$$

Controlled NOT (CNOT)

Control: 1, 0 NOT target

Control: 1, 1 NOT target

Control: 1, 0 NOT target

Control: 1, 1 NOT target

Control: 1, 0 NOT target

Control: 1, 1 NOT target

Control: 1, 0 NOT target

Control: 1, 1 NOT target

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Reversible computing a computation is reversible if it is always possible to uniquely recover the input, given the output

NOT

AND

REVERSIBLE

NOT REVERSIBLE

any (generally irreversible) computation can be transformed into a reversible computation

each gate in a finite family of gates can be made reversible by adding some additional input and output wires if necessary

the Toffoli gate is a reversible three-bit gate that flips the third bit, if and only if the first two bits are both in state 1

the reversible AND (Toffoli) gate is a generalization of the CNOT gate, where there are two bits controlling whether the NOT is applied to the third bit

if we have an irreversible circuit with depth T and space S

a reversible version uses a total of $O(S + ST)$

might require fewer than twice the number of gates as a non-reversible circuit for implementing f

the description of the universe given by quantum mechanics differs in fundamental ways from the classical description

input, workspace / junk, copy

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light and electrons have a lot in common

Wave-Particle duality

electromagnetic radiation

two oscillating mutually perpendicular fields

defines light color perceived

wavelength (λ)

crest

frequency (ν)

amplitude

speed of light

$$v = \frac{c}{\lambda}$$

Wave crests

Particle beam

wave particles 'blend'

crest

frequency (ν)

amplitude

speed of light

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information must be stored in some physical medium and manipulated by some physical process

the laws of physics ultimately dictate the capabilities of any information-processing machine

a probabilistic Turing machine is implicitly a classical machine

we could build it out of relatively large physical components, and all the aspects of its behavior relevant to its performing a computation would be accurately predicted by the laws of classical physics

attempting to use computers to simulate systems whose behavior is explicitly quantum mechanical, we do not seem to be able to do so efficiently

important problems can be solved efficiently on a quantum computer but no efficient solution is known on a probabilistic Turing machine

if we do not measure which path the photon is in, immediately after it passes through the first beam splitter, then its state remains

now, if the photon is allowed to pass through the second beam splitter (before making any measurement), its new state vector is

the second beam splitter has caused the two paths in superposition to interfere, resulting in cancellation of the '0' path

quantum mechanics

estado de polarização circular (à esquerda ou à direita)

amplitude no eixo x é 1

amplitude no eixo y é i ($e^{i\pi/2}$)

interação com a matéria

simetria e conservação do momento angular

transições atômicas

Teoria: Computação tem que ser possível

Prática: Computadores quânticos existentes ainda não são universais e escaláveis

Máquinas de Turing resolvem todos os problemas algorítmicos

Tese de Church-Turing (TCT)

Funções computáveis por dispositivos fisicamente realizáveis = Funções computáveis por MTs

para qualquer problema computacional

se existe um programa em C que resolve uma instância de tamanho n deste problema em $f(n)$ passos, então existe uma MT que resolve a mesma instância em $O(f(n)^k)$ passos

se existe uma MT que resolve uma instância de tamanho n do problema em $f(n)$ passos, então existe um programa em C que resolve a mesma instância em $O(f(n)^k)$ passos

existem problemas para os quais algoritmos quânticos são exponencialmente mais rápidos que algoritmos clássicos conhecidos

no nível mais fundamental da natureza, o meio físico parece fazer diferença na eficiência da computação de alguns problemas

objeto físico (x): modelo matemático

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